EPOCH ERP — SYSTEM AUDIT REPORT

vs. Deltek Costpoint

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Classification: Internal — Ownership Review

EXECUTIVE SUMMARY

EPOCH v8 is a custom-built, full-stack manufacturing ERP built for a small-to-mid-size composites manufacturer. It is operationally strong — p tracking, traveler execution, barcode-driven traceability, and P2 serialized item control are all well-implemented. However, EPOCH carries sign areas:

1. Financial backbone is fragmented. There is no enforced project-based accounting. Two parallel accounting constructs exist (Cost Accou Shadow Layer) that do not share a data model and neither is automatically populated from operations.
2. Multiple sources of truth exist across critical domains. The orders domain has four deprecated/transitional fields in active production table tables exist with no enforced canonical linkage at the database level.
3. Compliance readiness is not system-enforced. Audit trails exist but are domain-specific and inconsistent. There is no unified, immutable e CMMC controls are structurally present.

EPOCH's strongest competitive position is in manufacturing operations intelligence, AI-native direction (DONNA, pattern awareness layer), an orchestration. These advantages are real and should be protected while closing the financial and compliance gaps.

PHASE 1 — DOMAIN INVENTORY

Domains Identified

Domain	Source of Truth Table(s)	Key Fields	FK Relationshi
Orders (P1)	all_orders	order_id, status, status_id, current_department, current_department_id, customer_id, is_paid	status_id !' orde current_departm
Shipping	p2_shipping (P2), shipment_accounting_snapshots	tracking, carrier, ship date	Partial — snaps
AR / Invoicing	ar_invoices, ar_invoice_lines, ar_payments, ar_payment_allocations, credit_memos	total_amount, status, due_date, balance	credit_memo_id ar_invoices
Customers (P1)	customers	customer_id (text PK), balance fields	No FK to orders text
Customers (P2)	p2_customers	id, name, canonical key	Separate table; service
Vendors	vendors, vendor_scope_items, vendor_scope_groups	name, score, evaluation	vendor_scope_i
Inventory	inventory_items, inventory_balances	ag_part_number, quantity_on_hand	inventory_balan integer FK)
Production / Manufacturing	production_orders, travelers, traveler_steps, part_routings, robust_boms	order_id, current_department, traveler steps	production_orde match, not FK)
Timekeeping	punch_events, time_clock_entries, work_buckets	canonical_id, punch_time, job_id	punch_events.jo

Projects (P2 Pipeline)	projects, project_steps, project_activity_log	project_code, status, current_stage, po_id	po_id !’ p2_purc employees
Quality / Nonconformance	nonconformance_records, p2_nonconforming_dispositions, p2_serialized_items	status, disposition_type	disposition_id !’
Payments (P1)	payments, bulk_payment_batches	order_id, payment_type, payment_amount	order_id text ma
Cost Accounting	account_categories, accounts, monthly_account_entries, allocation_rules, allocation_results	account number, category, monthly amount	accounts !’ acco
GL / Journal (Shadow)	chart_of_accounts, journal_entries, journal_lines	account_name, account_type, debit_amount, credit_amount	journal_lines !’ c journal_entries (
Estimating	estimating_rfqs, estimating_rfq_parts, estimating_bom_lines, estimating_pricing_snapshots	part details, pricing, tooling	Internal FKs with
Purchase Orders (P2)	p2_purchase_orders, p2_purchase_order_items	po_number, status, customer_id	customer_id !’ p

PHASE 2 — SYSTEM OF TRUTH ANALYSIS

Domain Truth Map

Domain	Single Source of Truth?	Duplicate State Fields	DB-Level FK Enforcement
Orders	& þ PARTIAL	YES — status (text, marked legacy) + status_id (FK); current_department (text) + current_department_id (FK); priority_score (deprecated) + manual_priority_override; is_paid (boolean) duplicates payments table	PARTIAL — status_id and dept_id status/current_department text fie written
Customers	'L NO	Two tables: customers (P1) and p2_customers (P2), linked only via application-layer customerKey service	No cross-table FK
Shipping	& þ PARTIAL	shipment_accounting_snapshots is a snapshot layer — can diverge from live shipping records	sales_order_id reference not enfo
AR / Invoicing	' YES	Clean model — balance derived at query time from invoice - allocations - credits	FKs enforced on all AR tables
Inventory	& þ PARTIAL	inventory_items and inventory_balances joined by ag_part_number string (not integer FK)	NO integer FK between items and
Production	& þ PARTIAL	production_orders.order_id is a text field matching all_orders.order_id (not an integer FK)	NO enforced FK between product and all_orders
Timekeeping	' YES	punch_events is single source; time_clock_entries is legacy/parallel	job_id !’ production_orders.id (opt
Projects	' YES	Projects pipeline is clean, steps linked by UUID FKs	FKs enforced within project modu
Cost Accounting	& þ PARTIAL	Parallel to GL Shadow Layer — two separate COA constructs with no link	accounts !’ account_categories (C chart_of_accounts

Payments	'L NO	all_orders.is_paid, all_orders.payment_amount duplicate payments table	No FK between payments.order_i all_orders
Quality	' YES	p2_serialized_items is clean source for P2 QC; nonconformance_records for P1	FKs enforced in P2 dispositions
Vendors	' YES	vendors is clean	FKs enforced in scope tables

Critical Duplicate Fields (by table)

`all_orders`:

- status (text, legacy) + status_id (integer FK) — both written in production
- current_department (text) + current_department_id (integer FK) — both written
- priority_score (integer, marked DEPRECATED in code) + manual_priority_override (integer, newer model)
- is_paid + payment_amount + payment_date — duplicate of payments table

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PHASE 3 — ACCOUNTING READINESS

Category	Status	Notes
General Ledger Structure	PARTIAL	Two constructs exist: chart_of_accounts + journal_entries (GL shadow layer, wire payments only) and account_categories + accou accounting module, manually entered). They do not share a data model and neither is auto-populated from operations.
Project-Based Accounting	FAIL	The projects table is a pipeline tracker (RFQ !’ Quote !’ PO !’ Production). It has no WBS hierarchy, no budget fields, no cost-to-date are tied to a project code.
Cost Pools (Labor, Overhead, G&A)	PARTIAL	allocation_rules table exists with source/target accounts and allocation method/value. monthly_account_entries holds monthly actu and is disconnected from actual labor transactions, purchase orders, or shipment costs.
Revenue Recognition Logic	FAIL	No automated revenue recognition. shipment_accounting_snapshots captures shipment data at fulfillment time for manual QuickBo source code itself as "disposable, migratable."
Audit Traceability	PARTIAL	Domain-specific audit logs exist: admin_audit_log, p2_shipping_audit_log, receipt_audit_log, customer_satisfaction_audit_log, met schema_change_log. There is no unified financial audit trail that traces a dollar from order !’ invoice !’ payment !’ GL entry.

PHASE 4 — COMPLIANCE GAP ANALYSIS (vs. Costpoint)

Compliance Area	Costpoint	EPOCH	Gap
DCAA Compliance	Native — cost accounting, timekeeping, and billing are integrated and auditable	Not applicable — no cost pool !” labor !” billing linkage	CRITICAL: burdened c
FAR / CAS Alignment	Built-in — contract types (FFP, T&M, Cost+), cost principle enforcement	None — no contract type modeling	CRITICAL:
Audit Trail Completeness	Unified, immutable transaction log across all modules	Fragmented — domain-specific logs, no unified financial event log	HIGH: No s lineage
Role-Based Access Control	Granular — field-level and function-level RBAC	Basic — three roles (ADMIN, EMPLOYEE, OWNER) plus hardcoded username restrictions (e.g., glennj for accounting)	MEDIUM: U brittle at sc
Data Immutability	Financial records locked after posting	AR invoices lock after POSTED/SENT/VOID/PAID status — good. P1 payments are soft-mutable. Orders can be patched freely.	MEDIUM: I guarantee

Government Reporting	300+ standard reports	Custom reporting only — no standard government report formats	HIGH: No
CMMC Alignment	FedRAMP-capable environment	p2_shipping_audit_log referenced as "CMMC/DCAA compliant" in code comments but not structurally enforced	HIGH: Con
Contract Management	Full lifecycle: CLIN/SLIN, multi-contract projects, funding ceilings	None — project module tracks pipeline stages, not contract terms or funding	CRITICAL:

Explicit Gap List

1. No project accounting hierarchy (Project !’ Task !’ Cost)
2. No WBS or funding ceiling tracking on any project
3. Two disconnected accounting constructs with no reconciliation path
4. Timekeeping labor captured but not burdened or allocated to cost pools
5. Revenue recognition is manual and export-driven (QuickBooks dependency)
6. No standard government reporting formats
7. Dual customer master tables with application-layer-only linkage
8. Dual status fields on all_orders create audit ambiguity
9. payments table and all_orders payment fields not enforced to stay in sync
10. No unified immutable financial event log
11. Username-based access control is not scalable or auditable
12. production_orders.order_id links to all_orders via text match, not FK

PHASE 5 — ARCHITECTURE COMPARISON (EPOCH vs. Costpoint)

Dimension	Costpoint	EPOCH
Data Model Structure	Single unified relational model — all modules share one schema	15,339-line schema file — broad coverage, but contains deprecated/transitional fields in active tables and two separate accounting constructs
Module Integration	All modules tied back to Project !’ Contract !’ Cost Pool	Modules are route-based but loosely coupled — production orders reference all_orders via text join, not integer FK
Workflow Orchestration	Rigid, compliance-driven workflow steps	Strong — traveler execution, P2 pipeline steps, checklist instances, barcode-driven routing
Reporting / BI	Unified data model enables 300+ out-of-box reports	Domain-specific queries, shipment accounting snapshots, live metrics registry, executive rundown
AI Integration	Dela AI — query and automate within Costpoint data	Pattern Awareness Layer, DONNA agent, voice notes, codebase chat, document intelligence
Schema Governance	Managed by Deltek	Governance module exists: schema_change_log, migration_guard, schema_policy.ts, mutation logger — well-designed

PHASE 6 — RISK ASSESSMENT

Critical Risks (will break at scale)

Risk	Description
Dual customer master explosion	customers (P1) and p2_customers (P2) will diverge further as P2 grows. No DB-level constraint prevents the same custome incompatible records.

Dual status fields causing data inconsistency	all_orders.status (text) and status_id (FK) are both written. If they diverge, query results will be non-deterministic depending used.
Text-based joins between production_orders and all_orders	production_orders.order_id is a text field. No FK constraint means production records can exist without a parent order, and a order IDs would silently break production history.
Payments table vs. all_orders payment fields	is_paid, payment_amount, payment_date stored on orders AND in the payments table. A failed write to one leaves the syste inconsistent financial state.

Financial Risks (misstated revenue)

Risk	Description
QuickBooks dependency for revenue	Revenue recognition flows entirely through manual export of shipment_accounting_snapshots to QuickBooks. If a snapshot is missed, skippe is misstated with no system-level detection.
Two COA systems never reconcile	chart_of_accounts (journal entry shadow layer) and account_categories/accounts (cost accounting module) have no structural connection. A invisible to the cost accounting module and vice versa.
No project-level cost tracking	Costs (labor, materials, overhead) are not tied to projects. It is impossible to derive true project profitability from current system data.

Compliance Risks (audit failure points)

Risk	Description
Labor hours not burdened	Timekeeping captures punch events and calculates hours, but applies no burden rates. DCAA requires burdened labor costs allocated to
No funding ceiling enforcement	If a government contract has a funding ceiling, EPOCH has no mechanism to detect or prevent overbilling.
Username-based access control	Accounting features are gated with hardcoded username arrays (e.g., const AUTHORIZED_USERS = ['glennj']). This cannot be audited, i personnel changes.
Fragmented audit logs	Audit trails span 7+ separate log tables with different schemas. A DCAA auditor requesting a transaction log would receive incomplete, no

Operational Risks (duplicate systems, manual fixes)

Risk	Description
P1 / P2 boundary ambiguity	EPOCH has a clear P1 (custom stock/consumer orders) and P2 (contract manufacturing, serialized) split in the UI and routes, b payments, and orders exist in both universes without a unified view.
Governance layer not connected to all write paths	The mutation logger and governance routes are active, but many inline route handlers in routes/index.ts bypass the mutation lo
accountingPrep described as disposable in its own source	The shipment accounting snapshot module is self-described as "disposable, migratable." This means the only automated financ replaced, creating a future migration risk with no replacement yet defined.

PHASE 7 — PRIORITIZED ROADMAP

TIER 1 — MUST FIX (System Viability)

#	Problem	Impact	Recommended Fix
T1-1	Dual status fields on all_orders (status text + status_id FK both written)	Data inconsistency, query non-determinism, audit ambiguity	Complete migration to status_id FK. Add DB-level trigg legacy column.
T1-2	Text-based join between production_orders and all_orders	Cannot enforce referential integrity; silent orphan risk at scale	Add integer FK production_orders.all_order_id !’ all_or

T1-3	Dual customer master (customers + p2_customers)	Same customer has two records; no DB-level canonical link	Add a canonical_customers master table. Both custom customerKey service becomes a read path, not the onl
T1-4	Payment data duplicated on all_orders	Financial inconsistency if either write path fails	Remove is_paid, payment_amount, payment_date fron payments table. Add a view or join for display if needed
T1-5	inventory_balances joins inventory_items via string part number	Cannot enforce referential integrity on inventory	Add integer FK inventory_balances.inventory_item_id !

TIER 2 — ENTERPRISE READINESS

#	Problem	Impact	Recommended Fix
T2-1	Two disconnected accounting constructs	Financial data siloed; impossible to produce a unified P&L	Merge into a single Chart of Accounts. Cost accounting module accounts b entries post to these accounts.
T2-2	No project-based cost tracking	Cannot derive project profitability or support DCAA cost allocation	Extend projects with budget fields. Add cost_transactions table that ties lab production-touching transactions should carry project_id.
T2-3	Labor not burdened	Hours captured but costs not calculated per DCAA requirements	Add burden rate configuration table. Extend labor summary service to appl cost transactions.
T2-4	Fragmented audit logs	Cannot produce unified financial audit trail	Create a single financial_audit_log table. All financial events (invoices, payr write one canonical row here in addition to domain-specific logs.
T2-5	Username-based access control	Brittle, unauditable, breaks on personnel changes	Replace with a permissions table. Feature access is tied to employee roles source code.
T2-6	Revenue recognition is manual (QuickBooks export)	Missed shipments = misstated revenue; no system-level detection	Auto-generate journal entries at shipment time from within EPOCH. Reven becomes a secondary output, not the primary record.

TIER 3 — COMPETITIVE ADVANTAGE (Where EPOCH beats Costpoint)

#	Opportunity	What to Build
T3-1	Domain Truth Inspector (already started)	Expand into a live, scheduled data integrity monitor that flags constraint violations, orphan records, and field conflicts
T3-2	AI-driven project cost intelligence	DONNA + Pattern Awareness Layer surfaces budget burn anomalies, predicts cost overruns, and flags unusual labo No off-the-shelf ERP does this well.
T3-3	Unified financial event stream as AI query target	Once Tier 2 produces a canonical financial_audit_log, DONNA can answer natural-language financial queries ("Wha without custom reports.
T3-4	Live compliance readiness score	A real-time DCAA/compliance readiness scorecard driven by actual data — missing burden rates, unlinked transacti score to ownership. This alone would be a market differentiator.

SUMMARY SCORECARD

Area	Rating	Notes
Manufacturing Operations	' STRONG	Best-in-class shop floor, traveler execution, barcode traceability
Order Management	& þ PARTIAL	Functional but carrying legacy dual-field debt
Inventory	& þ PARTIAL	String joins, no FK integrity between items and balances
AR / Invoicing	' GOOD	Clean model, proper balance derivation at query time
Payments	'L WEAK	Duplicated across two locations, no FK enforcement

Customer Master	'L WEAK	Two tables, no enforced canonical link at DB level
Cost Accounting	& þ PARTIAL	Two disconnected accounting constructs; manual entry only
GL / Journal Entries	& þ PARTIAL	Wire payments only; not transaction-complete
Project Accounting	'L FAIL	Pipeline tracker only; no financial project structure
Labor / Timekeeping	& þ PARTIAL	Hours captured; no burdening, no project cost allocation
Compliance (DCAA/FAR)	'L FAIL	Aspirational only; not structurally enforced
Audit Trails	& þ PARTIAL	Domain-specific only; no unified financial event log
RBAC / Access Control	& þ PARTIAL	Three-role system; username hardcoding is a meaningful gap
AI / Automation	' STRONG	Ahead of Costpoint — DONNA, Pattern Awareness, voice notes
Schema Governance	' GOOD	Governance layer, migration guard, schema policy — well-designed

This report was produced via read-only inspection of the EPOCH schema (server/schema.ts, 15,339 lines), all route files, service layer, gover
pages. No code was modified. All findings are based on verified schema definitions, route logic, source comments, and architectural patterns.